

Team UpembaPark

Data Card template

Intelligent Complaint Classification via Transformer Architectures

Dataset: CFPB Consumer Complaint Database filtered(Kaggle)- 2011-2024

Input vs. Output	• What are users inputting (e.g., audio, text, images)? • What are the outputs or labels your model will predict or generate?	Your response: Input: consumer complaint narratives (free text). Output: predict Product category
Classification problem	• What is the classification task, if any? • What categories or classes are you working with, and why?	Your response: Supervised multi-class text classification — narrative in, Product category out. Uses CFPB's existing Product taxonomy (e.g., credit cards, checking/savings, mortgages, student loans) — chosen for a clear, feasible target for a transformer model.
Labeling	• Where do your labels come from — domain experts, community members, crowd workers, AI-generated sources? • How will you check for consistency and accuracy?	Your response: Labels come directly from CFPB's own Product field, not team- or AI-generated. Quality check: review missing labels, duplicates, class distribution, and inconsistencies before training.
Data creation	• How will you collect your data?	Your response: The dataset was pulled from Kaggle and originally sourced from CFPB's site: 5M+ total complaints filtered to 2M+ from December 2011- August 2024
Web collection ethics	• If collecting from the web, how will you address consent, copyright, privacy, and legal/ethical considerations?	Your response: The dataset is obtained from Kaggle but originates from publicly released CFPB Consumer complaint database. We respect privacy safeguards applied by CFPB to published narratives and will not attempt to rewrite or reproduce complaint content
Bias and representation	• What groups may be underrepresented or excluded? • How will you improve diversity, fairness, and representation?	Your response: Only consumers who filed a complaint with an available consumer narrative are included; 65% of all complaints(3.8M) WERE DROPPED for lacking one. Mitigations: check class imbalance, use stratified splitting or class-weighted training, evaluate with per-category precision/recall/F1 rather than overall accuracy alone.

Values and community alignment	• How does your dataset design reflect the values from your Challenge 1 problem statement? • How will it benefit and remain accountable to the intended community?	Your response: Privacy: rely on CFPB's existing consent and scrubbing safeguards. Fairness: check imbalance and evaluate performance across categories, not just in aggregate. Human oversight: model assists classification/organization only — not an independent high-stakes decision-maker.
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